

advance and short weaker stocks during declines, which is very much in tune with relative strength momentum investing.

On the quantitative side of things, beginning in the late 1920s, Arnold Bernhard, founder of the Value Line Investment Survey, successfully used relative strength price momentum in conjunction with earnings growth momentum. According to the Value Line website, Group 1 stocks are those ranked highest in performance over the past year and that had accelerating earnings growth. From 1965 through 2012, Group 1 stocks had an average annual gain of 12.9% before dividends, while the S&P 500 gained 6.1%. Group 5 stocks had a -9.8% annual loss. Dividing a stock's latest 10-week average relative performance by its 52-week average relative price is the price momentum factor still used today by Value Line.

H. M. Gartley developed momentum-based relative velocity ratings in the 1920s. The Dow theorist Robert Rhea (1932) subsequently published these ratings in his book *The Dow Theory*. Gartley (1945) himself wrote an article titled "Relative Velocity Statistics: Their Application in Portfolio Analysis" for the *Financial Analysts Journal*. In that article Gartley wrote, "In addition to the usual valuation methods applied to stock, analysts should consider its velocity. The velocity statistic is a technical factor in the stock's price volatility that measures the percentage rise and fall of a stock against an average." This was yet another way to express relative strength price momentum.

In his 1935 book *Profits in the Stock Market*, Gartley introduced the world to trend-following moving averages. Bernhardt and Gartley were both early pioneers of quantitative, rules-based momentum strategies.

The first truly scientific study and published academic paper on momentum was by Alfred Cowles III and Herbert E. Jones (1937). Cowles was a prominent economist who founded the Cowles Foundation for Economic Research, initially at the University of Chicago and now at Yale. There were no computers back then, so Cowles and Jones painstakingly hand-compiled stock performance statistics from 1920 through 1935. This was a remarkable accomplishment at that time. Cowles and Jones found that the strongest stocks during the preceding year had a very strong tendency to remain strong during the following year. In their own words, "Taking one year as the unit of measurement for the period 1920 to 1935, the tendency is very pronounced for stocks which have exceeded the